VNAV Lab4: UAV Control and Racing

Boyu Huang, 24020036026; Class: International Computer Science, Class 1; Group 8

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Summary

I connected the Lab3 controller to the Lab4 trajectory generator, tested hovering at the first gate, and recorded one complete course run. The planner started at the measured state, passed through 12 gate targets, and sent position, velocity, acceleration, and yaw references at 50Hz. Time-aligned bag data gave an execution-only position RMSE of 5.64cm and a maximum error of 8.44cm. Height was the main source of position error: the vehicle averaged 5.47cm below its reference. This run confirms that the programs work together. It does not establish the cause of the height offset or the benefit of faster timing.

1 Introduction

Lab3 follows a reference. Lab4 decides what that reference should be over time. Hovering needs one fixed target; flying through gates needs a timed route. Passing the next gate position directly to the controller would not specify how to move between gates.

I first checked the coordinate and control chain at the first gate, then ran the full trajectory. The planner supplies position and heading. The controller uses the measured state to correct deviations and adjust rotor speeds. This repeated correction is closed-loop control. [1,2]

The work covered Part0 and Parts1.1-1.3, two individual theory exercises, and the video, screenshots, and message records needed to inspect the result. Reliable completion was the goal. No repeated search for the fastest settings was carried out.

2 Experimental Set-up

Table 1 lists the configuration used for this recording. The reported statistics all come from that run, rather than a mixture of parameter settings.

Table 1. Environment and operating parameters

Item	Setting	Purpose
Operating system and ROS	Ubuntu 20.04.6; ROS Noetic	Run and connect the programs
Simulation and code	TESSE Unity Lab4; VNAV2021	Supply the vehicle, course, and starter code
Controller gains	kx=2; kv=3; kr=3; komega=1	Position, velocity, attitude, and angular-rate feedback
Time multiplier	time_scale=2.0	Double the estimated segment durations
Reference update rate	50Hz	Update approximately every 0.02s
Timing estimates	v_max=8.0; a_max=4.0	Estimate segment times; not enforced flight limits
Position and yaw dimensions	3 and 1	Plan XYZ and heading separately
Polynomial coefficients	N=10	At most degree nine per segment

ROS is the framework used to exchange messages between programs. A program is called a node, and a named message channel is a topic. The bridge publishes the simulated state on /current_state. The gate publisher supplies target poses. The trajectory node combines those targets with the current vehicle state and publishes /desired_state. The controller then sends /rotor_speed_cmds back to the simulator.

2.1 Trajectory and controller implementation

The completed code is organized into four parts. The main files are simple_traj_planner.cpp and trajectory_generation_node.cpp.

Table 2. Implementation and design choices

Part	Implementation	Reason
Part0	Use the first gate as a fixed target with zero desired velocity and acceleration	Check coordinates and basic control before moving flight
Part1.1	Read current position and yaw; check that the values are valid	Avoid planning from an uninitialized state
Part1.2	Add start and gate constraints; optimize position and yaw separately	Reach targets while coordinating heading
Part1.3	Sample by elapsed time and publish the reference state	Convert a complete trajectory into a current control target

Yaw is the horizontal heading angle. Its numerical value wraps at plus or minus 180 degrees. A change from 179 to -179 degrees requires only a 2-degree rotation. The implementation uses atan2(sin delta_psi, cos delta_psi) to select the short change between consecutive headings and accumulates these changes into a continuous sequence. This prevents an unnecessary near-full turn.

Position has three components, whereas yaw has one. The optimizations are separate but use the same segment times, so the position and heading references reach each gate at the same planned instant. The initial position and yaw come from the current state. Start and end conditions use makeStartOrEnd; intermediate gates

constrain position and yaw. Roll and pitch are handled by the controller rather than prescribed as separate route trajectories.

Minimum snap minimizes the integral of the squared fourth derivative of position. In practical terms, it discourages abrupt changes in the reference motion. The program estimates segment durations, multiplies them by the time scale, and solves for polynomial coefficients with a linear optimizer. It does not directly minimize rotor energy and does not perform online obstacle replanning. [3]

At each update, sampling time is the current ROS time minus the trajectory start time, clamped between zero and the total duration. Pose is placed in transforms, linear velocity in velocities, and linear acceleration in accelerations. Yaw is converted into a quaternion, a four-number representation of three-dimensional orientation. The final reference is held after the trajectory ends. The inherited Lab3 simplification omits desired angular-rate feedforward, meaning no additional advance compensation for reference turning rate was used.

2.2 Procedure

Basic control was checked before the complete recorded run.

Table 3. Procedure and checks

Step	Action and purpose	Outcome
1 Prepare the workspace	Resolve dependencies and build	All 20 packages compiled
2 Test the first gate	Start the simulator, bridge, and fixed target	Stable holding near the first gate
3 Change mode	Stop the fixed target, reset, and restart the follower	Avoid competing reference publishers
4 Record the run	Start the follower and recorder before publishing gates	Save continuous video and a bag file
5 Check evidence	Replay the video and inspect state topics	Playable video and both required state streams

Resetting Unity alone does not clear the trajectory stored in the follower. A repeat run therefore also requires restarting that node. Repeated gate messages do not regenerate an existing trajectory; this prevents normal flight from being repeatedly restarted.

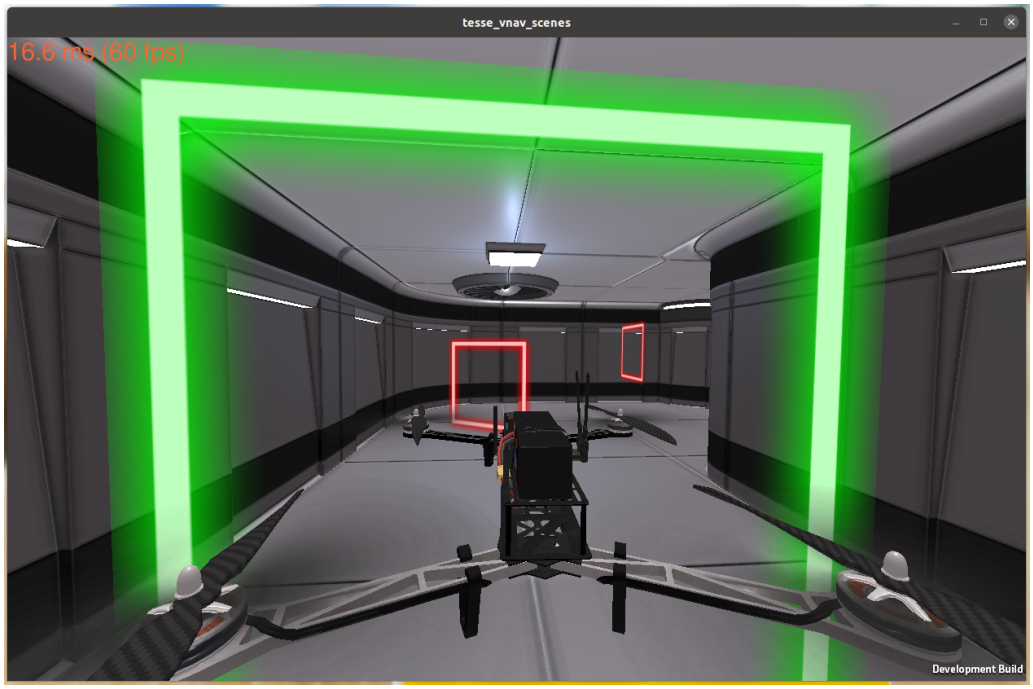


Figure 1. First-gate hovering test with a fixed reference

2.3 Recording and data processing

The analyzed recording is run_20261022_000059. A bag file stores ROS messages with their recording times. This file contains 40,534 messages, including 8,365 actual-state messages and 7,836 desired-state messages. It also retains the simulation clock, gate targets, and rotor commands.

The two state streams contain different message counts, so subtracting entries by array index would be inappropriate. Desired positions were linearly interpolated to actual-state recording times: adjacent reference values were used to estimate the value at an intermediate instant. Yaw was unwrapped before interpolation. The trajectory origin was estimated from recording time minus elapsed trajectory time during advancing samples. The final hold was excluded, leaving 6,708 execution samples.

Position error is actual position minus desired position. The three-dimensional root mean square error, or RMSE, is $\mathrm{sqrt}(\mathrm{mean}(ex^2+ey^2+ez^2))$. It summarizes the errors while giving greater weight to larger deviations. The signed mean on each axis was also calculated to distinguish persistent offset from fluctuations around zero. [4]

2.4 Reasons for the design choices

Starting from the measured state avoids attaching the vehicle to a route based on an arbitrary default position. The fixed-target test checks the basic chain before a time-varying reference is introduced. Table 4 summarizes the main choices.

Table 4. Design choices and their limits

Choice	Reason	Limit in this experiment
Measured position and yaw at the start	Match the route to the current state	Valid state messages are still required
Separate position and yaw, shared timing	Coordinate position and heading at each gate	Roll and pitch are not independently prescribed
Time multiplier of 2.0	Allow more execution time for tracking	No measured comparison with other multipliers
Generate once after valid gate input	Repeated messages do not restart the route	A new run requires restarting the follower
Clamp time at the endpoint	Avoid out-of-range sampling and hold the final target	Completing the route does not stop the controller

For the same geometric curve, stretching time by a factor of two halves velocity and quarters acceleration. This explains why timing affects tracking demands. It is a scaling relationship, not a measured improvement from a second run: that comparison was not performed.

2.5 Checking the message chain

An open simulator window alone does not show that the programs are connected. Actual state provides feedback, desired state provides the reference, and rotor commands provide the control action.

Table 5. Recorded topics and their roles

Topic	Messages	Role
/current_state	8365	Actual position, orientation, and velocity
/desired_state	7836	Reference state for the current instant
/desired_traj_vertices	785	Ordered gate targets
/rotor_speed_cmds	15352	Controller output to the rotor interface
/clock	8196	Simulation time for timed execution

Different counts do not directly prove packet loss. Topics can have different update rates and active intervals. For error analysis, the important checks are overlapping coverage and a common time base, rather than equal array lengths.

3 Results

The measurements below separate the hovering snapshot from statistics over the complete execution interval.

3.1 First-gate hovering

Table 6 contains one hovering snapshot. Horizontal positions were nearly identical, while the vehicle was approximately 3.96cm below the target. This single reading is not the average over the racing trajectory.

Table 6. One first-gate position snapshot, in metres

Axis	Desired	Actual	Actual minus desired
X	0.4226201	0.4226182	-0.0000019
Y	3.1652320	3.1652690	+0.0000370
Z	2.6700000	2.6304210	-0.0395790

3.2 Complete course run

The saved continuous video records one complete course run. Figure 2 shows the scene after the final gate. There were 12 gate targets, excluding the current position added as the start vertex.

Table 7. Results during trajectory execution

Metric	Value
Execution samples	6,708
Three-dimensional position RMSE	0.056426m
Mean / median position error	0.055555 / 0.056864m
95th-percentile / maximum position error	0.068054 / 0.084428m
X / Y / Z RMSE	0.007171 / 0.006392 / 0.055602m
X / Y / Z signed mean error	-0.000647 / +0.000435 / -0.054671m
Yaw RMSE	2.2922 degrees
Maximum / mean actual speed	1.6292 / 0.7125m/s
Accumulated sampled path length	95.570m

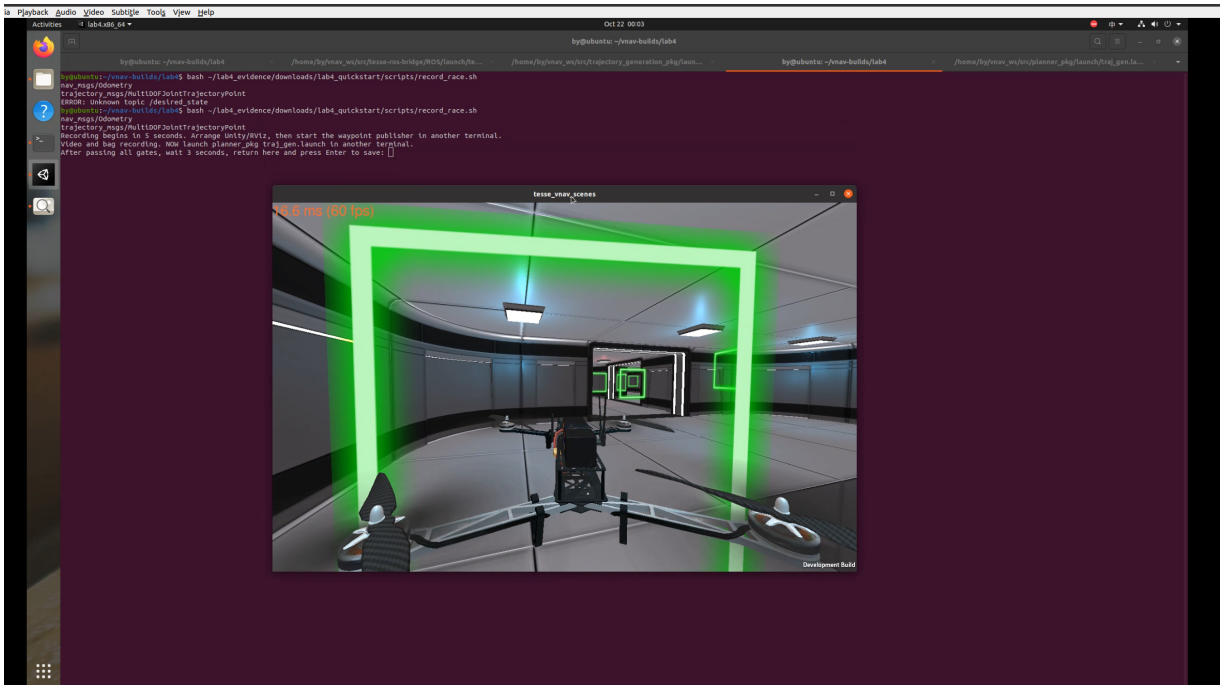


Figure 2. Scene after the final gate; passage is assessed from the continuous video

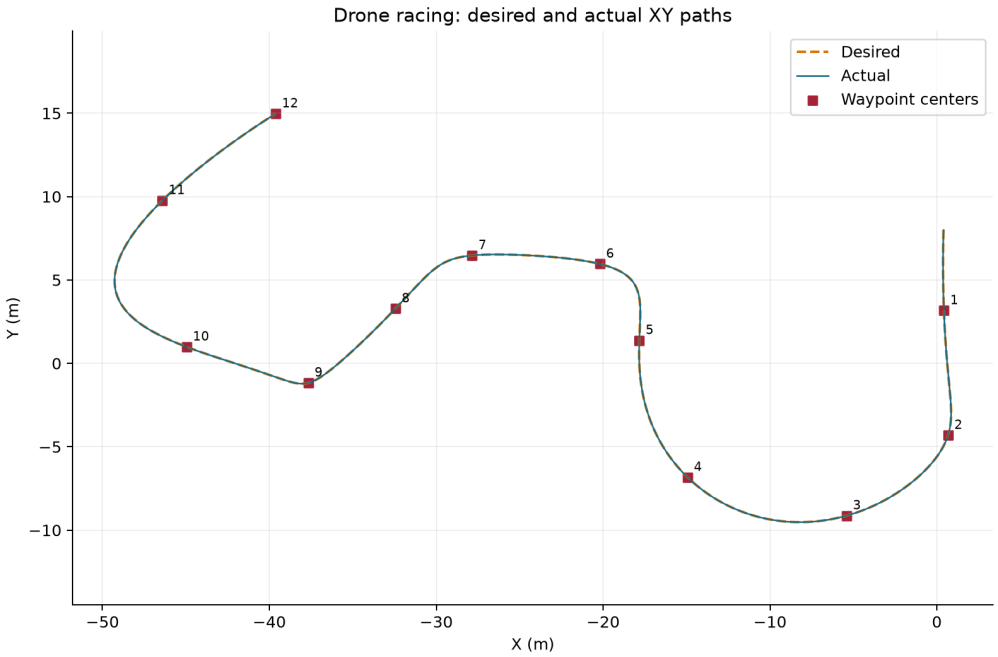


Figure 3. Desired and actual trajectories in the horizontal plane

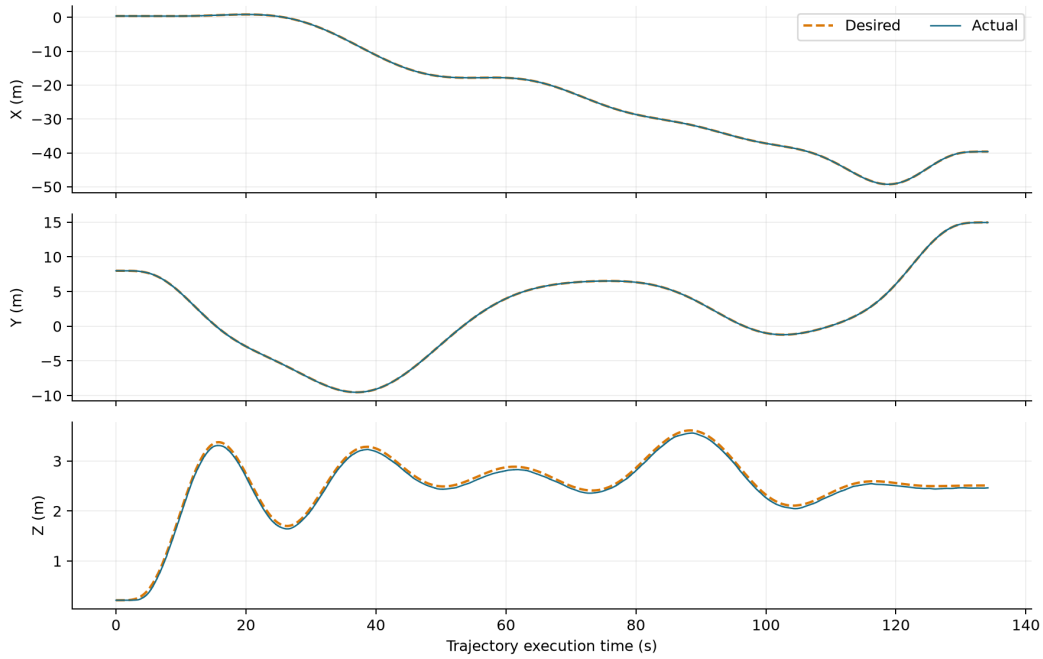


Figure 4. Desired and actual XYZ positions over time

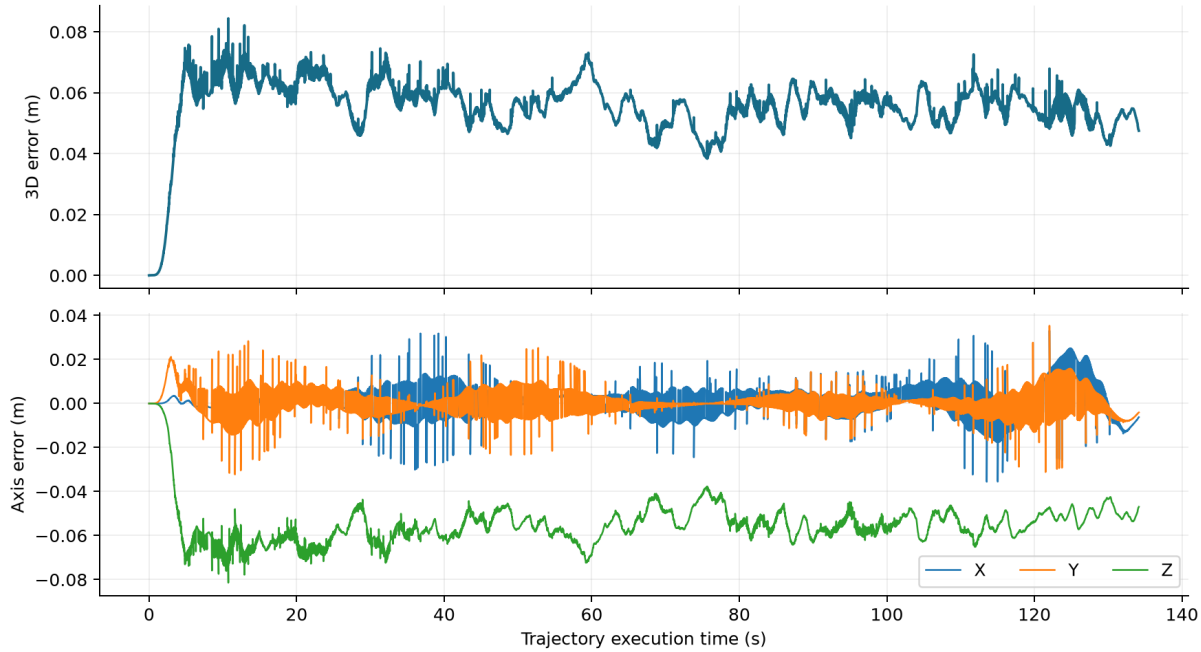


Figure 5. Three-dimensional and per-axis position errors



Figure 6. Per-axis position RMSE over the execution interval

3.3 Duration measurements

The recording windows and the planned execution interval are different. Their durations should not be treated as interchangeable.

Table 8. Recording and reference durations

Item	Duration	Meaning
Bag recording	167.28s	Preparation, execution, and final holding
Screen recording	160.75s	Playback length of the video file
Reference trajectory	134.17s	Planned trajectory execution interval
Independently adjudicated race time	[to be completed]	No separate timing record was provided

4 Discussion

Completing the route, tracking it accurately, and racing quickly are different outcomes. The recording supports the first two for this run; the remaining questions need separate tests.

4.1 Tracking quality and vertical offset

Horizontal RMSE was approximately 0.72cm in X and 0.64cm in Y, compared with 5.56cm in Z. The signed vertical mean was -5.47cm, close in magnitude to the vertical RMSE. This indicates that a persistent downward offset accounts for much of the vertical error, rather than purely positive and negative fluctuations. The hovering snapshot also shows a downward offset. However, a snapshot and an execution-wide average are different measurements; their difference does not establish that the offset grew over time.

The 95th-percentile error was approximately 6.81cm and the maximum was 8.44cm. Together with the mean, these values describe the distribution for this run. They do not establish performance under different parameters, on another course, or on a physical vehicle. No controlled isolation of controller gains or model terms was conducted, so the specific cause of the height offset remains unresolved.

4.2 Connection to the theory exercises

The single-segment exercise fixes time and displacement, then minimizes the integral of squared velocity. Squaring gives speed peaks a larger cost, so constant velocity is optimal under the original position constraints. This chosen objective is not a model of battery use. With only P(0)=0 and P(1)=1, the optimum is P(t)=t and J=1. Allowing quadratic or cubic terms does not reduce the cost. Adding P'(1)=-2 changes the feasible trajectories: the quadratic optimum is $4t-3t^2$ with J=4, while the cubic optimum is $t/4+9t^2/2-15t^3/4$ with J=17/8. Additional polynomial freedom is useful only when it can satisfy constraints and improve the objective.

For one-dimensional minimum snap with fixed segment times, prescribed waypoint positions, and fixed low-order endpoint derivatives, each segment can be represented by a degree-seven polynomial. For k segments there are 8k coefficients: 2k position equations, six endpoint derivative equations, 3(k-1) low-order continuity equations, and 3(k-1) natural optimality conditions. Natural conditions arise from optimization; they should not all be counted as independently imposed constraints. For three segments, 24 coefficients correspond to 18 explicit constraints and six natural conditions.

The implementation uses N=10, meaning ten coefficients and at most degree nine. Its boundary settings and implementation choices differ from the degree-seven theoretical setting. N should therefore not simply be changed to seven. The English files deliverable_1.pdf and deliverable_2.pdf contain the full models, solutions, and checks.

4.3 Measurement and experimental limitations

Errors are relative to the state reported by the simulator, not to an independent external measurement system. Alignment by bag recording time cannot fully separate communication delay from controller error. RMSE over the entire overlapping interval was 0.054428m; it includes final holding and is not substituted for execution-only RMSE. No backward clock steps were detected, but this does not eliminate other sampling or communication effects.

A screenshot describes only one instant. Likewise, the closest distance to a gate centre does not independently prove a correct passage. Complete-course assessment relies on the continuous video and the experimenter's run check; numerical analysis supplements that evidence. Accumulated sampled path length is also not the exact geometric length of the course.

Only one complete run was supplied, without separate adjudicated timing. The value 134.17s is therefore reported as reference duration rather than an independently verified race score. A time multiplier of 2.0 allowed more time for tracking, but no controlled comparison against 1.0 or other values was made, so its benefit cannot be quantified here.

4.4 Debugging and further work

A missing gflags_catkin dependency was resolved by adding its source. A stale catkin_simple path was resolved by backing up the old build and devel spaces and rebuilding. After the NLopt download connection was refused, a verified local archive was used; all 20 packages then compiled. The first recording attempt could not find /desired_state. Once the topic was available, the second attempt saved successfully. These were dependency, cache, download, and startup-order problems rather than a single common failure.

Further work should first hold the route and time multiplier fixed, change one controller or model parameter at a time, and compare signed height error, maximum error, and completion. After reliable tracking is established, segment times could be shortened gradually and assessed over repeated runs. These are proposed experiments, not completed results.

4.5 Separating vertical offset from variation

The three-dimensional RMSE alone does not identify the dominant axis. Per-axis statistics allow a horizontal combined error to be calculated. The vertical mean square can then be separated into squared mean offset and mean-centred variation.

Table 9. Additional diagnostics derived from the same statistics

Diagnostic	Value	Calculation and meaning
Combined horizontal RMSE	0.96cm	sqrt($RMSE_X^2$+$RMSE_Y^2$)
Vertical share of 3D mean squared error	97.10%	$RMSE_Z^2$ / $RMSE_3D^2$; height dominates the position error
Squared mean offset as a share of vertical mean square	96.68%	$mean(eZ)^2$ / mean(eZ^2); persistent offset dominates
Mean-centred vertical RMS variation	1.01cm	sqrt(mean(eZ^2)-$mean(eZ)^2$)

These are further calculations on the same run, not new measurements. Mean squared errors add across axes; RMSE values themselves do not. A large offset share also does not justify simply raising the target by the measured offset as a general fix. That might mask the symptom without testing its cause.

4.6 A controlled follow-up plan

Investigating the error and seeking a faster run should be separate tasks. First repeat the baseline, then change one factor at a time. Otherwise, an improvement cannot be linked to a particular change.

Table 10. Proposed checks; not yet performed

Order	Change or check	Compare
1 Repeat the baseline	Keep route, gains, and time multiplier fixed	Completion, signed height error, maximum position error
2 Investigate height	Check coordinates, gravity compensation, and model conventions separately	Whether the height offset changes consistently
3 Adjust one controller gain	Hold other parameters fixed	Offset, variation, and completion together
4 Shorten segment times gradually	Reduce the multiplier only after reliable tracking	Success rate over repeated runs and separately measured completion time

5 Conclusions

The four parts now form a complete sequence: hold a fixed target, read the actual start, plan position and yaw, and publish the reference by elapsed time. One saved run completed the course. Execution-only position RMSE was 5.64cm, with a maximum error of 8.44cm. Horizontal errors were small; a persistent downward offset dominated the position error. Its cause has not been isolated.

The next step is to keep the route and time scale fixed while investigating height error. Faster settings should then be compared over repeated runs. Code, original video, the bag, and screenshots are archived. Both individual theory deliverables are in English. The required public Drive/Dropbox media links remain [to be completed].

References

[1] MIT VNAV. Lab 4: UAV Control and Racing. https://vnav.mit.edu/labs_2023/lab4/exercises.html

[2] MIT-SPARK. VNAV-labs, VNAV2021, lab4. https://github.com/MIT-SPARK/VNAV-labs/tree/VNAV2021/lab4

[3] ETH Zurich ASL. mav_trajectory_generation. https://github.com/ethz-asl/mav_trajectory_generation

[4] Experiment recording run_20261022_000059; repository files data/analysis/metrics.json and analyze_run.py; theory derivations in deliverable_1.pdf and deliverable_2.pdf.

Execution, screenshots, and video came from the author's Ubuntu environment. AI assistance was used for code preparation, theory checks, offline analysis, and document preparation.

Appendix A. Required core code with annotations

The excerpts below follow the archived run implementation, with explanatory comments added. Template instructions, blank lines, and progress logging are omitted. The valid-state guard in A.3 omits its warning text. These are excerpts, not standalone compilable files; complete context is retained in code/.

Table A1. Tasks, functions, and verification material

Task	Core function	Verification material
Part0	trajCB	Hovering image and reference/actual positions
Part1.1	onCurrentState	Current-state topic and validity checks
Part1.2	generateOptimizedTrajectory	Waypoint and trajectory plots; yaw unwrapping
Part1.3	publishDesiredState	Desired-state messages and execution error plot

A.1 Part0

```
// Use the first gate; the callback has already checked for an empty array.
const auto& pose = traj_msg.poses.front();
trajectory_msgs::MultiDOFJointTrajectoryPoint point;
geometry_msgs::Transform transform;
transform.translation.x = pose.position.x;
transform.translation.y = pose.position.y;
transform.translation.z = pose.position.z;
transform.rotation = pose.orientation;
// Append the pose to the message array; a local variable alone is not sent.
point.transforms.push_back(transform);
// A default Twist has zero linear and angular velocity: a stationary target.
point.velocities.emplace_back();
// Add a zero acceleration entry so the expected array is populated.
point.accelerations.emplace_back();
point.time_from_start = ros::Duration(0.0);
// Publish the fixed reference for the controller.
desired_state_pub.publish(point);
// Build the world-to-desired_pose transform to inspect the target.
tf::Transform desired;
desired.setOrigin(tf::Vector3(pose.position.x, pose.position.y, pose.position.z));
tf::Quaternion q;
tf::quaternionMsgToTF(pose.orientation, q);
// Normalize the quaternion before broadcasting the target transform.
q.normalize();
desired.setRotation(q);
br->sendTransform(tf::StampedTransform(desired, ros::Time::now(), "world", "desired_pose"));
```

A.2 Part1.1

```
// Read world-frame XYZ from odometry for the trajectory start.
x << cur_state.pose.pose.position.x, cur_state.pose.pose.position.y,
     cur_state.pose.pose.position.z;
// Extract current heading from the orientation quaternion.
current_yaw = tf::getYaw(cur_state.pose.pose.orientation);
// Accept only finite values; keep NaN and infinity out of the optimizer.
have_current = x.allFinite() && std::isfinite(current_yaw);
```

A.3 Part1.2

```
// Wait for a valid state; the original warning text is omitted here.
if (!have_current) return;
// Do not replace an existing route when gate messages repeat.
if (!trajectory.empty()) return;
// Use three position dimensions; optimize yaw separately in one dimension.
const int D = 3;  // dimension of each vertex in the trajectory
mav_trajectory_generation::Vertex start_position(D), end_position(D);
mav_trajectory_generation::Vertex::Vector vertices;
mav_trajectory_generation::Vertex start_yaw(1), end_yaw(1);
mav_trajectory_generation::Vertex::Vector yaw_vertices;
using namespace mav_trajectory_generation::derivative_order;
// Set the actual start and the library-provided endpoint derivative conditions.
start_position.makeStartOrEnd(x, SNAP);
vertices.push_back(start_position);
Eigen::VectorXd initial_yaw(1); initial_yaw << current_yaw;
// Start heading at the measured yaw rather than an arbitrary zero.
start_yaw.makeStartOrEnd(initial_yaw, SNAP);
yaw_vertices.push_back(start_yaw);
double last_yaw = current_yaw;
// Add position and heading for each gate using the same gate index.
for (auto i = 0; i < poseArray.poses.size(); ++i) {
// Read the pose of the gate at the current index.
  const auto& pose = poseArray.poses[i];
  Eigen::Vector3d position(pose.position.x, pose.position.y, pose.position.z);
  mav_trajectory_generation::Vertex vertex(D);
// Use endpoint conditions at the final gate; fix values at internal gates.
  if (i + 1 == poseArray.poses.size())
    vertex.makeStartOrEnd(position, SNAP);
  else
    vertex.addConstraint(POSITION, position);
  vertices.push_back(vertex);
  double yaw = tf::getYaw(pose.orientation);
// Accumulate the short angle difference to avoid a long turn across +/-pi.
  yaw = last_yaw + std::atan2(std::sin(yaw - last_yaw), std::cos(yaw - last_yaw));
  mav_trajectory_generation::Vertex yaw_vertex(1);
  Eigen::VectorXd yaw_value(1);
  yaw_value << yaw;
// Use endpoint conditions at the final gate; fix values at internal gates.
  if (i + 1 == poseArray.poses.size())
    yaw_vertex.makeStartOrEnd(yaw_value, SNAP);
  else
    yaw_vertex.addConstraint(ORIENTATION, yaw);
  yaw_vertices.push_back(yaw_vertex);
  last_yaw = yaw;
}
```

A.3.1 Segment timing and optimization

```
std::vector<double> segment_times;
const double v_max = 8.0;
const double a_max = 4.0;
// Estimate segment times; these parameters are not hard speed constraints.
segment_times = estimateSegmentTimes(vertices, v_max, a_max);
// The recorded multiplier is 2.0, allowing more tracking time.
for (auto& duration : segment_times) duration *= time_scale;
// N counts coefficients: ten coefficients allow degree nine.
const int N = 10;
mav_trajectory_generation::PolynomialOptimization<N> opt(D);
// Set up the minimum-snap position problem with fixed segment times.
opt.setupFromVertices(vertices, segment_times, SNAP);
opt.solveLinear();
mav_trajectory_generation::PolynomialOptimization<N> yaw_opt(1);
// Use the same timing for yaw so heading and position remain synchronized.
yaw_opt.setupFromVertices(yaw_vertices, segment_times, SNAP);
yaw_opt.solveLinear();
mav_trajectory_generation::Segment::Vector segments;
opt.getTrajectory(&trajectory);
yaw_opt.getTrajectory(&yaw_trajectory);
// Record the start time after solving; publication uses this time origin.
trajectoryStartTime = ros::Time::now();
```

A.4 Part1.3

```
trajectory_msgs::MultiDOFJointTrajectoryPoint next_point;
// Sample by elapsed time rather than repeatedly sampling the start.
double sampling_time = std::max(0.0, (ros::Time::now() - trajectoryStartTime).toSec());
// Clamp at the final time to avoid sampling beyond the trajectory.
if (sampling_time > trajectory.getMaxTime())
  sampling_time = trajectory.getMaxTime();
next_point.time_from_start = ros::Duration(sampling_time);
using namespace mav_trajectory_generation::derivative_order;
// Evaluate position, velocity, acceleration, and yaw at the same instant.
Eigen::Vector3d des_position = trajectory.evaluate(sampling_time, POSITION);
Eigen::Vector3d des_velocity = trajectory.evaluate(sampling_time, VELOCITY);
Eigen::Vector3d des_accel =
    trajectory.evaluate(sampling_time, ACCELERATION);
Eigen::VectorXd des_orientation =
    yaw_trajectory.evaluate(sampling_time, ORIENTATION);
geometry_msgs::Transform transform;
transform.translation.x = des_position.x();
transform.translation.y = des_position.y();
transform.translation.z = des_position.z();
// Convert desired yaw to the quaternion required by the pose message.
transform.rotation = tf::createQuaternionMsgFromYaw(des_orientation(0));
geometry_msgs::Twist velocity, acceleration;
velocity.linear.x = des_velocity.x();
velocity.linear.y = des_velocity.y();
velocity.linear.z = des_velocity.z();
acceleration.linear.x = des_accel.x();
acceleration.linear.y = des_accel.y();
acceleration.linear.z = des_accel.z();
// Populate all three arrays before publishing; angular-rate feedforward is unset.
next_point.transforms.push_back(transform);
next_point.velocities.push_back(velocity);
next_point.accelerations.push_back(acceleration);
desiredStatePub.publish(next_point);
```

Appendix B. Experimental screenshots archived on GitHub

Figures 1 and 2 use original screenshots 04 and 07. The six remaining screenshots below come from this experiment in data/raw/screenshots and retain their original content, matching the GitHub archive.

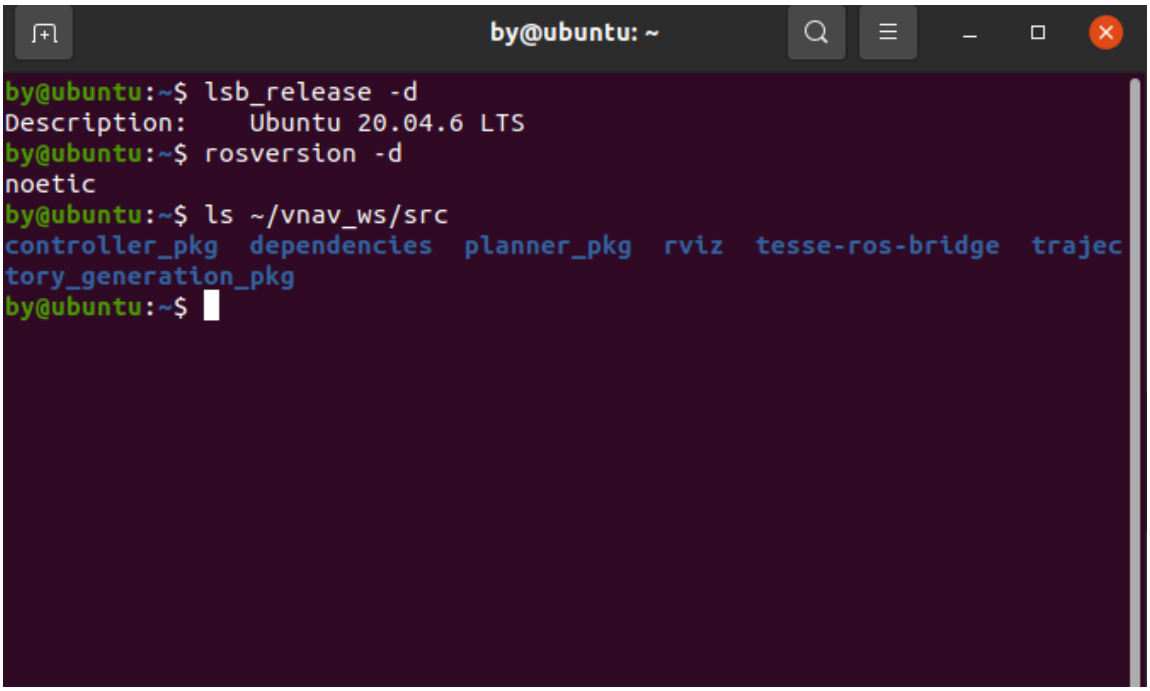


Figure 7. Environment and workspace; Section 2 (original screenshot 01)

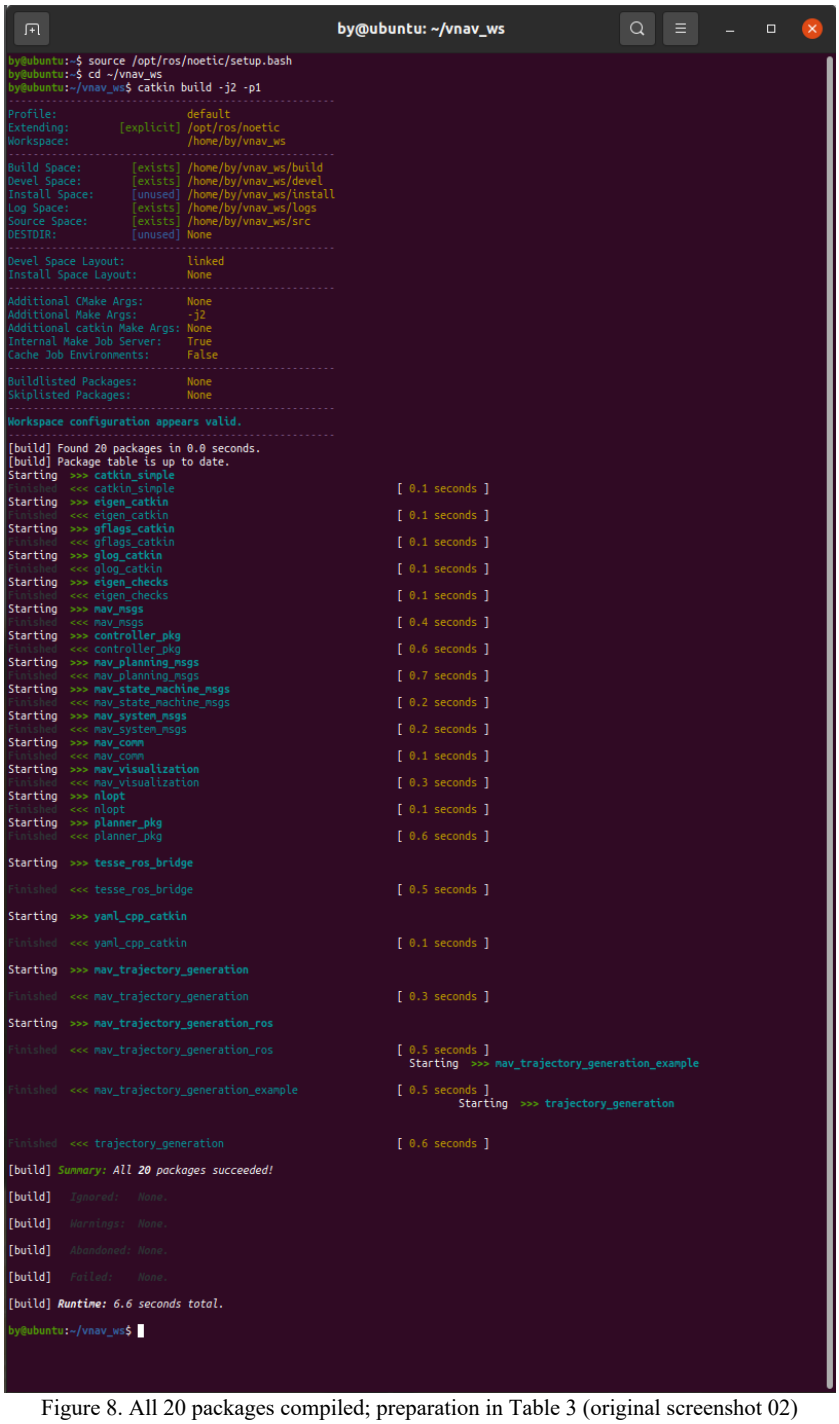


Figure 8. All 20 packages compiled; preparation in Table 3 (original screenshot 02)

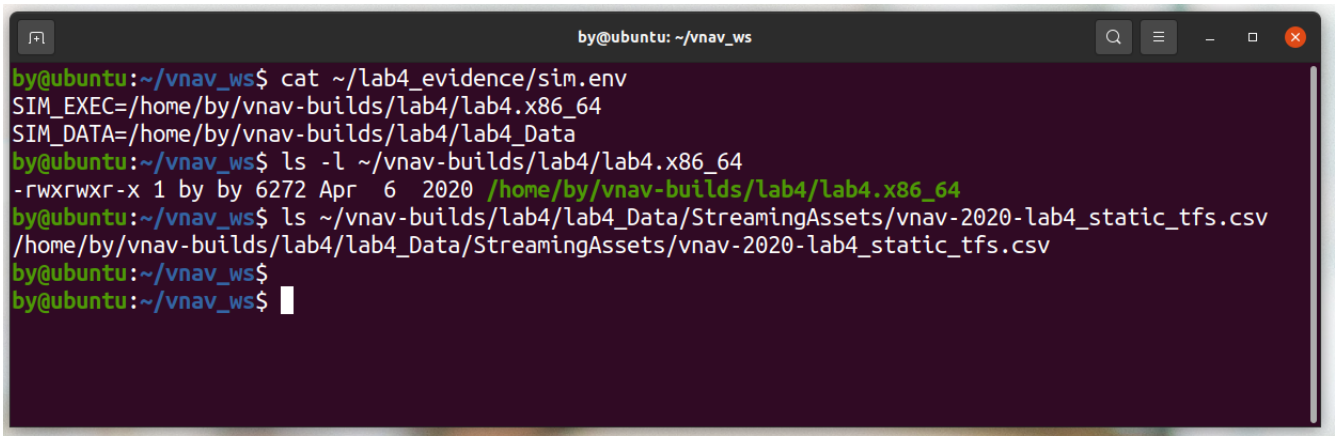


Figure 9. Simulator and gate-data paths; Section 2 (original screenshot 03)

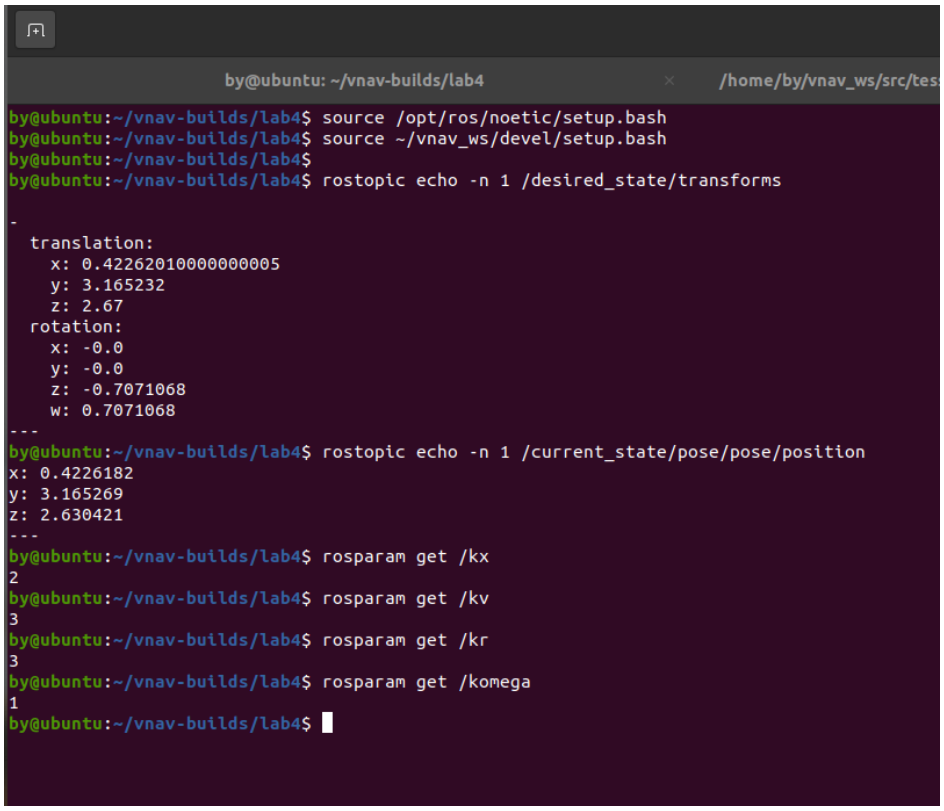


Figure 10. Hovering reference, actual position, and gains; Tables 6 and 1 (original screenshot 05)

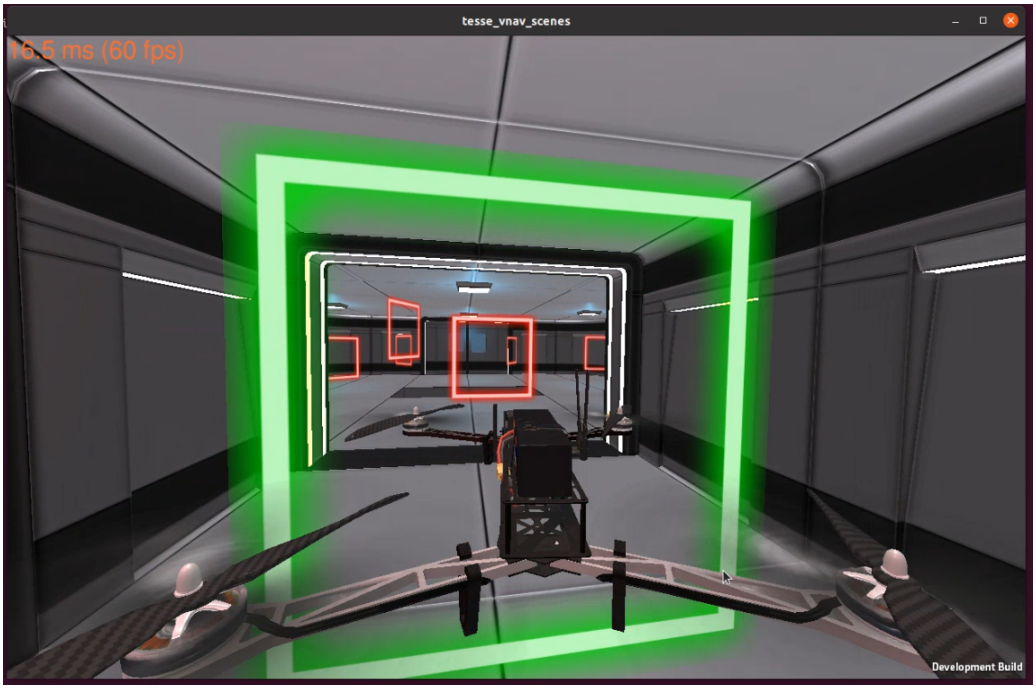


Figure 11. Scene during the race; Section 3.2 (original screenshot 06)

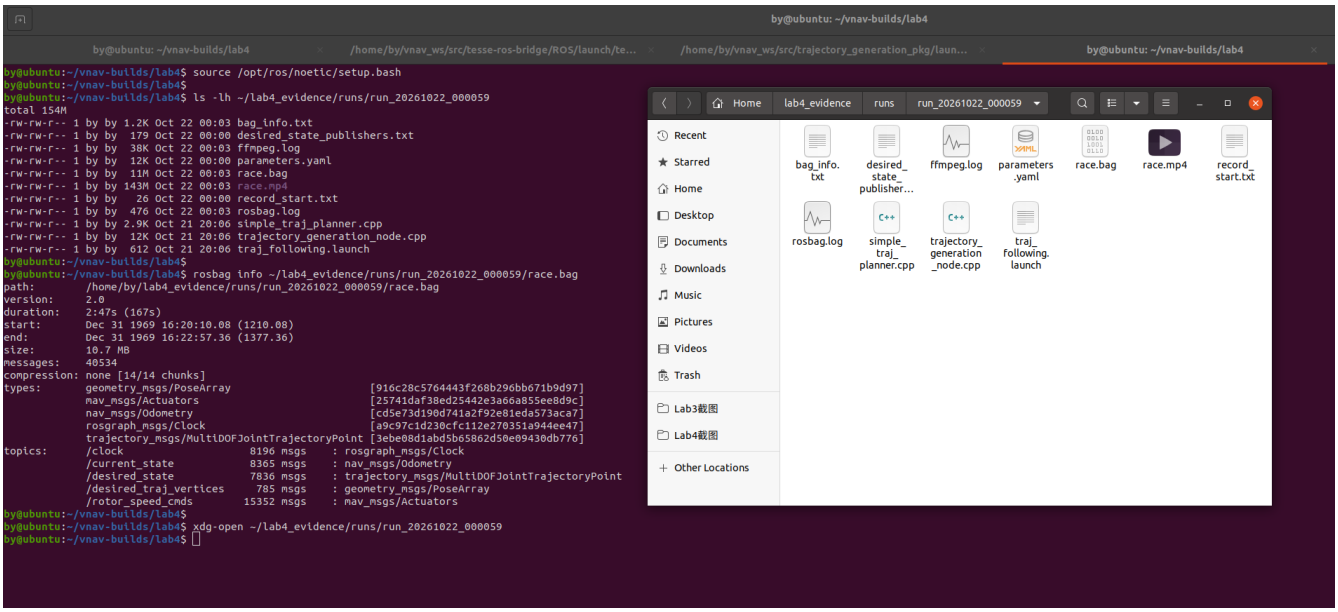


Figure 12. Recorded bag information and topic counts; Section 2.3 (original screenshot 08)

Appendix C. Three-dimensional trajectory

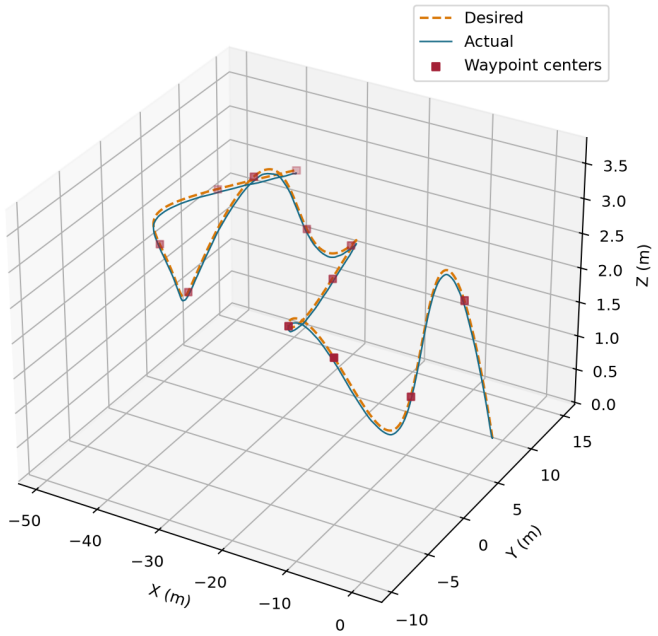


Figure 13. Three-dimensional paths and gate centres; useful for viewing height changes, not an independent gate-passage test